

ST3248 Notes

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INTRODUCTION TO STATISTICAL LEARNING

1.1 Foundations of Statistical Learning and Conditional Expectation

1.1.1 Core Concepts

Across very different applications, statistical learning asks us to extract structure from data. The presence or absence of a response distinguishes into two settings.

Definition 1.1 (Supervised Learning) For each observation, we observe predictors and a response variable.

- Regression: response is quantitative (e.g. price, income, temperature)
- Classification: response is categorical (e.g. spam/genuine, healthy/sick)

Definition 1.2 (Unsupervised Learning) We observe predictors, but no response variable.

- reduce dimension and visualise structure
- discover groups of similar observations

1.1.2 The Regression Function

Definition 1.3 (The Regression Function)

The function

$$m(x) := E[Y \mid X = x] \quad (1.1)$$

is called the regression function. It describes how the **expected value of Y changes with X** .

Object	Type	Interpretation
$E[Y \mid X = x]$	number	conditional mean at x
$m(\cdot)$	function	maps x to the conditional mean
$E[Y \mid X] = m(X)$	random variable	m evaluated at random X

Conditional Expectation

Conditional expectation behaves much like ordinary expectation, but with X treated as fixed/known.

- **Linearity:** $E[aY + bZ \mid X] = aE[Y \mid X] + bE[Z \mid X]$
- **Taking out what is known:** $E[g(X)Y \mid X] = g(X)E[Y \mid X]$
- **Independence:** If X and Y are independent, then $E[Y \mid X] = E[Y]$
- **Tower property:** $E(E[Y \mid X]) = E[Y]$ and $E(E[Y \mid Z, X] \mid X) = E(Y \mid X)$

Conditional Variance

The conditional mean

$$m(X) = E[Y \mid X]$$

describes the part of Y that can be predicted from X .

The variation that remains after observing X is measured by

$$\text{Var}(Y \mid X) = E[\{Y - E[Y \mid X]\}^2 \mid X]$$

Theorem 1.4 (Law of Total Variance)

$$\text{Var}(Y) = \mathbb{E}[\text{Var}(Y | X)] + \text{Var}(\mathbb{E}[Y | X]) \quad (1.2)$$

where $\text{Var}(Y)$ is the total variation, $\mathbb{E}[\text{Var}(Y | X)]$ is the variation remaining after observing X , and $\text{Var}(\mathbb{E}[Y | X])$ is the variation explained by X .

Example 1.5 (Toy Example)

Let $X \in \{0, 1, 2\}$ be the number of exclamation marks in an email, and let

$$Y = \begin{cases} 1, & \text{spam} \\ 0, & \text{genuine} \end{cases}$$

Suppose the joint distribution of (X, Y) is

	$x = 0$	$x = 1$	$x = 2$	$p_Y(y)$
$y = 0$	0.4	0.15	0.05	0.60
$y = 1$	0.05	0.10	0.25	0.40
$p_X(x)$	0.45	0.25	0.30	1

From the marginals,

$$\mathbb{E}[X] = 0.85, \quad \mathbb{E}[Y] = P(\text{spam}) = 0.40$$

Without looking at X , every email receives the same prediction $P(\text{spam}) = 0.40$. Hence, does knowing X improve this prediction? If X does not contain information about Y , then X and Y are independent and their joint probabilities would factorize. But, for example,

$$p(0, 0) = 0.40 \neq p_X(0)p_Y(0) = 0.45(0.60) = 0.27$$

Hence, X and Y are dependent, i.e. the distribution of Y changes with X . The covariance describes the direction of their linear association:

$$\begin{aligned} \mathbb{E}[XY] &= (1)(1)(0.10) + 2(1)(0.25) = 0.60 \\ \text{Cov}(X, Y) &= 0.60 - (0.85)(0.40) = 0.26 > 0 \end{aligned}$$

Since the covariance is positive, X and Y move in the same direction: more exclamation marks are associated with a higher probability of an email being spam.

Since Y is binary, its regression function is the conditional probability of spam:

$$m(x) = \mathbb{E}[Y | X = x] = P(\text{spam} | X = x)$$

Using the joint distribution,

$$m(0) = \frac{0.05}{0.45} = \frac{1}{9}, \quad m(1) = \frac{0.10}{0.25} = \frac{2}{5}, \quad m(2) = \frac{0.25}{0.30} = \frac{5}{6}$$

Thus, conditioning on X turns one baseline prediction into predictions tailored to the observed characteristic X of the email.

Knowing X reduces our uncertainty about Y since without observing X , Y is Bernoulli with success probability 0.40, so

$$\text{Var}(Y) = 0.40(1 - 0.40) = 0.24$$

Given $X = x$, Y is Bernoulli with success probability $m(x)$, so

$$\text{Var}(Y | X = x) = m(x)\{1 - m(x)\}$$

Hence, averaging over X ,

$$E[\text{Var}(Y | X)] = 0.45 \left(\frac{8}{81} \right) + 0.25 \left(\frac{6}{25} \right) + 0.30 \left(\frac{5}{36} \right) \approx 0.146$$

By law of total variance (1.2), total variance is 0.24, variation remaining after observing X is 0.146, and the variation explained by X is 0.094. Knowing X reduces the average remaining uncertainty from 0.24 to 0.146.

Remark 1.1.1 (1) We learn how Y depends on X and use this relationship to predict new observations. (2) Without X , the natural prediction is $E[Y]$, but after observing X , we can use $E[Y | X]$. (3) The regression function captures the predictable variation of Y , and informative predictors can improve prediction by explaining more of the variation in Y .

More predictors explains more variation

Suppose we have an additional predictor Z . Then

$$E[E[Y | X, Z] | X] = E[Y | X]$$

by the tower property.

Apply the law of total variance to

$$U = E[Y | X, Z]$$

Then,

$$\begin{aligned} \text{Var}(E[Y | X, Z]) &= \text{Var}(E[U | X]) + E[\text{Var}(U | X)] \\ &= \text{Var}(E[Y | X]) + E[\text{Var}(E(Y | X, Z) | X)] \end{aligned}$$

Since $E[\text{Var}(E(Y | X, Z) | X)] \geq 0$, therefore,

$$\text{Var}(E[Y | X]) \leq \text{Var}(E[Y | X, Z])$$

This means at the population level, additional predictors can only increase the variation explained by the regression function.

So far, the population regression function $m(x) = E[Y | X = x]$ is treated as if it were known. In practice, m is unknown and must be **estimated from finite data**. Using more predictors and fitting more flexible relationships can make this estimation problem harder.

- Bias: A model that is too simple may miss important structure
- Variance: A model that is too flexible may change substantially from one sample to another

1.2 The Regression Model

1.2.1 Model Definition and Variation

Let Y be a quantitative response and $X = (X_1, \dots, X_p)^\top$ a vector of predictors.

Definition 1.6 (Homoscedastic Regression Model)

We assume

$$Y = f(X) + \varepsilon \tag{1.3}$$

where

$$E[\varepsilon | X] = 0, \quad \text{Var}(\varepsilon | X) = \sigma^2 < \infty$$

The conditional mean-zero assumption implies

$$E[Y | X] = f(X) + E[\varepsilon | X] = f(X)$$

Hence

$$f(X) = m(X) = E[Y | X]$$

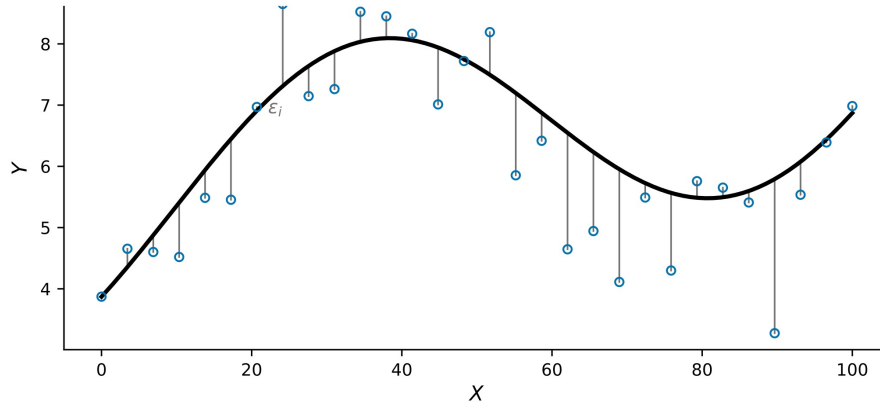
Thus under this model, the unknown function f is exactly the regression function (1.1).

Systematic Variation and Noise

The model $Y = f(X) + \varepsilon$ (1.3) separates the response into two parts:

1. $f(X)$ is the **systematic part** of Y : the conditional mean of Y given X , and hence the part predictable from X .
2. $\varepsilon = Y - f(X)$ is the **noise**: the remaining variation around the regression function. Since $E[\varepsilon | X] = 0$, it has no systematic component left once X is known.

$\sigma^2 = \text{Var}(\varepsilon | X)$ measures the amount of this remaining variation. Homoscedasticity means that it is the same for every value of X , while heteroscedastic data means $\text{Var}(\varepsilon | X) = \sigma^2(X)$.



- The black curve is $f(X) = E[Y | X = x]$, the systematic relationship.
- The vertical segments are $\varepsilon_i = y_i - f(x_i)$, the unsystematic noise terms.
- Given $X = x$, the observations vary around $f(x)$ with mean-zero noise and variance σ^2 .
- **We never observe f directly but rather the data scattered around it.**

1.2.2 Goals of Learning

Knowing f is useful for two quite different reasons.

Definition 1.7 (Prediction)

We want to *predict* the response Y from observed predictors X . We use

$$\hat{Y} := \hat{f}(X)$$

where what matters is how accurately $\hat{f}(X)$ predicts Y ; the form of $\hat{f}$ itself may be unimportant.

Definition 1.8 (Inference)

We want to *understand* how the response Y depends on the predictors X . We use

$$\hat{f}$$

to learn which predictors matter, in which direction, and how strongly. What matters is what $\hat{f}$ tells us about the relationship between X and Y .

	Prediction	Inference
main goal:	predict Y accurately	understand how Y depends on X
what matters:	$\hat{f}(x)$ is accurate	the structure of $\hat{f}$ is interpretable
complexity:	can be useful	can make interpretation harder

1.3 Evaluating Model Performance

We observe a training set $\mathcal{D} = \{(x_1, y_1), \dots, (x_n, y_n)\}$ where $y_i = f(x_i) + \varepsilon_i$. A **statistical learning method** uses these data to construct an estimate of f : $\mathcal{D} \mapsto \hat{f}_{\mathcal{D}}$. Because $\hat{f}$ is constructed from the data, different training samples will generally produce different fitted functions: $\mathcal{D}_1 \mapsto \hat{f}_{\mathcal{D}_1}$, $\mathcal{D}_2 \mapsto \hat{f}_{\mathcal{D}_2}$, $\dots$.

1.3.1 Measuring Accuracy

Suppose $\hat{f}$ has been fitted using the training data $\mathcal{D}$. Consider a new observation (X_0, Y_0) , drawn from the same population but not used to fit $\hat{f}$.

Definition 1.9 (Mean Squared Prediction Error)

The mean squared prediction error (MSPE) of the fitted function $\hat{f}$ is

$$E[\{Y_0 - \hat{f}(X_0)\}^2 \mid \mathcal{D}]$$

The expectation averages over new observations (X_0, Y_0) , while the fitted function $\hat{f}$ is held fixed (by conditioning on the data $\mathcal{D}$). A good fitted model predicts new observations accurately, i.e. has small MSPE.

The MSPE is a population quantity and is generally unknown. Two sample quantities are useful: training MSE to measure fit and test MSE to measure prediction.

Definition 1.10 (Training MSE)

Measures the fit to the **training data**, using the observations that were used to fit $\hat{f}$,

$$\text{MSE}_{\text{train}} = \frac{1}{n} \sum_{i=1}^n \{y_i - \hat{f}(x_i)\}^2$$

Definition 1.11 (Test MSE)

Estimates prediction error on **new data**, using new observations $(x_1^0, y_1^0), \dots, (x_m^0, y_m^0)$ not used to fit $\hat{f}$,

$$\text{MSE}_{\text{test}} = \frac{1}{m} \sum_{j=1}^m \{y_j^0 - \hat{f}(x_j^0)\}^2$$

1.3.2 Model Complexity and Overfitting

Different learning methods allow fitted functions of different complexity. For example, consider polynomials of increasing degree:

$$\mathcal{F}_1 \subset \mathcal{F}_2 \subset \mathcal{F}_3 \subset \dots, \quad \mathcal{F}_d = \{\beta_0 + \beta_1 x + \dots + \beta_d x^d\}$$

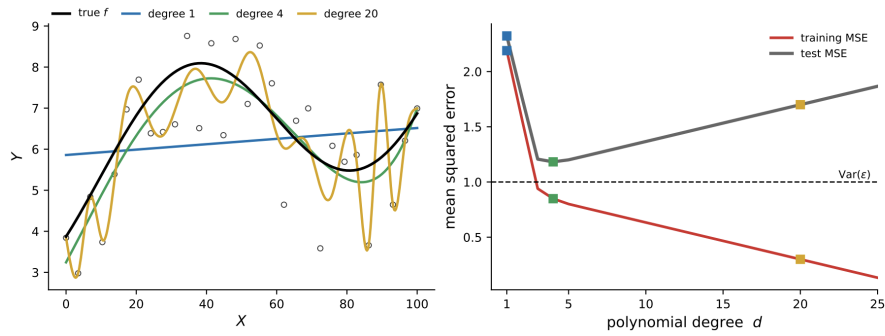
A larger class is **more flexible**: it allows $\hat{f}$ to take a wider variety of shapes. *Lower* flexibility would imply straighter lines while *higher* flexibility increases the complexity of the curves.

Greater flexibility makes it easier to fit the training data closely since as flexibility increases, we expect the fitted function to follow the training observations more closely. Hence, the training MSE should decrease. But the quantity we care about is the prediction error on new observations: **Does a smaller training MSE imply a smaller test MSE?**

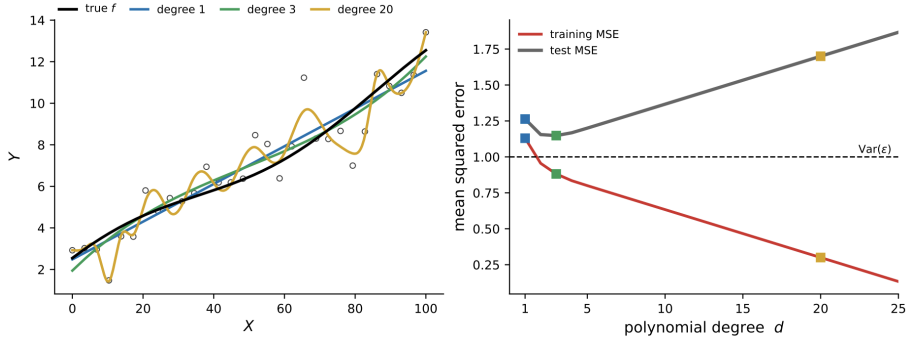
Training Error versus Test Error

To find out, we simulate data from a known regression function f . The left panel shows one training sample and three polynomial fits, where increasing flexibility lets the fitted curve follow the training data more closely. The right panel shows the training and test MSE as the flexibility increases. The horizontal dashed line is $\text{Var}(\varepsilon) = \sigma^2$, the irreducible error.

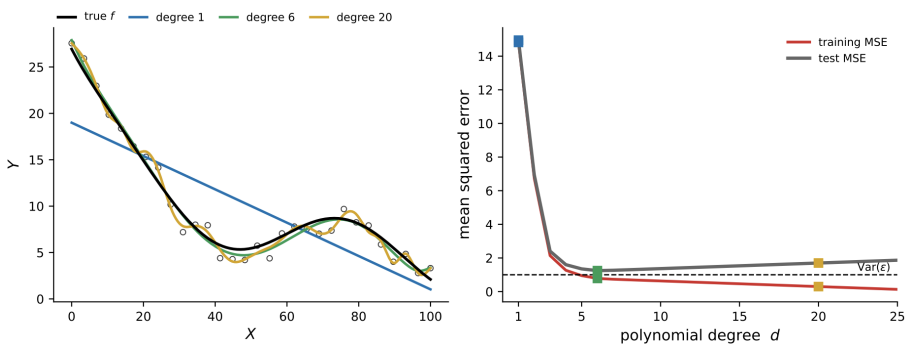
1. f is Non-Linear



2. f is Nearly Linear



3. f is Highly Non-Linear



The details differ across the three scenarios, but the qualitative message is stable:

- Training MSE decreases as flexibility increases.
- Test MSE is typically U-shaped.
- The optimal flexibility depends on the unknown true function f .
- Training MSE therefore cannot be trusted to choose flexibility: it tends to prefer models that are too flexible.

Overfitting

Definition 1.12 (Overfitting) Overfitting occurs when increasing flexibility improves fit to the training data but worsens prediction on new data.

Overfitting is seen on the right-hand side of the U-shaped test-error curve. The mechanism is already visible in the regression model:

$$y_i = f(x_i) + \varepsilon_i$$

where $f(x_i)$ is the signal and ε_i is the noise. A very flexible method adapts to both the signal and the sample-specific noise which does not repeat in a new sample. Overfitting means fitting features of the training data that do not generalise.

1.3.3 Optimal Predictors and Bias-Variance Trade-off

Suppose we could choose *any* function g to predict Y from X . Under squared-error loss, its mean squared prediction error is

$$E[\{Y - g(X)\}^2]$$

Theorem 1.13 (Error Decomposition)

Suppose

$$Y = f(X) + \varepsilon, \quad E[\varepsilon | X] = 0, \quad \text{Var}(\varepsilon | X) = \sigma^2$$

Then, for any predictor $g(X)$,

$$E[\{Y - g(X)\}^2] = E[\{f(X) - g(X)\}^2] + \sigma^2$$

Since $E[\{f(X) - g(X)\}^2] \geq 0$, the mean squared predictor error is minimised by setting $g(X)$ equal to

$$f(X) = E[Y | X]$$

Therefore, the regression function is the best possible predictor of Y under squared error.

Reducible and Irreducible Error

Hold the fitted function $\hat{f}$ fixed and consider prediction on a new observation. By theorem (??),

$$E[\{Y_0 - \hat{f}(X_0)\}^2 | \mathcal{D}] = E[\{f(X_0) - \hat{f}(X_0)\}^2 | \mathcal{D}] + \sigma^2$$

Definition 1.14 (Reducible Error)

Our fitted function $\hat{f}$ differs from the true regression function f . Better estimation of f can reduce this part.

Definition 1.15 (Irreducible Error)

Even if $\hat{f} = f$, the response still varies around $f(X)$ because of the noise ε . No prediction method can remove this part.

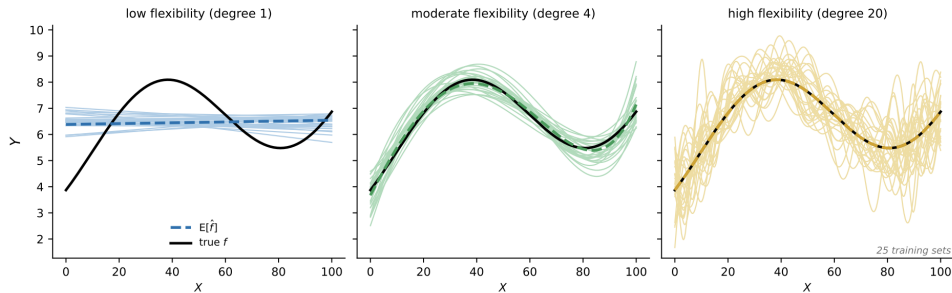
With the learning method fixed, $\hat{f}(x_0)$ is random because the training data are random.

$$\mathcal{D} \mapsto \hat{f}_{\mathcal{D}} \mapsto \hat{f}_{\mathcal{D}}(x_0)$$

If we repeatedly collect new training samples and refit the model, we obtain different predictions at the same point x_0 :

$$\hat{f}_{\mathcal{D}_1}(x_0), \quad \hat{f}_{\mathcal{D}_2}(x_0), \quad \hat{f}_{\mathcal{D}_3}(x_0), \quad \dots$$

Thus, $E[\hat{f}(x_0)]$ and $\text{Var}(\hat{f}(x_0))$ refer to repeated training samples. Estimation introduces a second source of randomness: the training sample itself.



Each thin curve is fitted from different training sets generated from the same population. The dashed curve is the average fit $E[\hat{f}]$; the black curve is the truth f . We see two entirely different things can go wrong at fixed point x_0 .

Definition 1.16 (Bias)

Measures systematic error

$$\text{Bias}(\hat{f}(x_0)) := \mathbb{E}[\hat{f}(x_0)] - f(x_0)$$

Systematically wrong: The method aims at the wrong place where $\mathbb{E}[\hat{f}(x_0)] \neq f(x_0)$. The fitted curves agree with one another but systematically miss f .

Definition 1.17 (Variance)

Measures sample-to-sample instability

$$\text{Var}(\hat{f}(x_0)) := \mathbb{E} \left[\left\{ \hat{f}(x_0) - \mathbb{E}[\hat{f}(x_0)] \right\}^2 \right]$$

Unstable: The method changes substantially from one training sample to another. The fits are close to f on average but individual fits vary widely.

Low flexibility tends to produce more bias and less variance while high flexibility tends to produce less bias and more variance.

Mean Squared Error DecompositionFix a test point x_0 and consider a new response

$$Y_0 = f(x_0) + \varepsilon_0, \quad \mathbb{E}[\varepsilon_0] = 0, \quad \text{Var}(\varepsilon_0) = \sigma^2$$

The training data used to construct $\hat{f}$ are independent of the new noise ε_0 . We study the **expected test error** at x_0 ,

$$\mathbb{E}[(Y_0 - \hat{f}(x_0))^2]$$

where the expectation averages over both (1) the random training data that determine $\hat{f}$, and (2) the new response Y_0 .

Theorem 1.18 (Mean Squared Error Decomposition)For a fixed test point x_0 ,

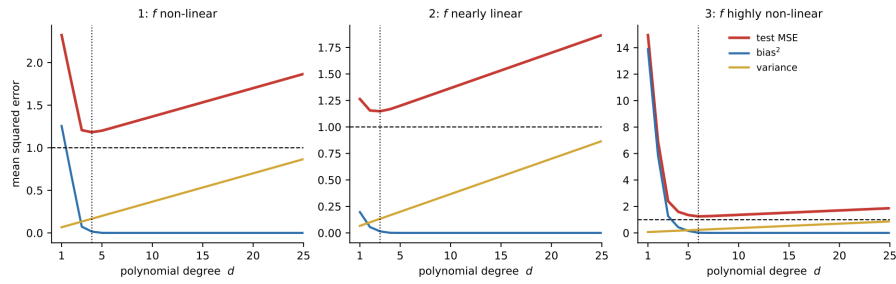
$$\mathbb{E}[(Y_0 - \hat{f}(x_0))^2] = \sigma^2 + \text{Bias}(\hat{f}(x_0))^2 + \text{Var}(\hat{f}(x_0))$$

- **Irreducible noise** (σ^2): randomness in Y_0 that no method can predict
- **Squared Bias** ($\text{Bias}(\hat{f}(x_0))^2$): systematic error of the learning model
- **Variance** ($\text{Var}(\hat{f}(x_0))$): instability cause by estimating f from a random finite sample

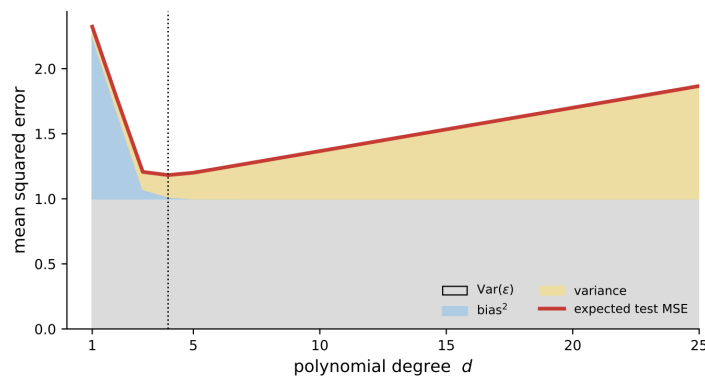
Term	What it measures	How can it change?
σ^2	noise in Y given X	cannot be reduced
$\text{Bias}(\hat{f}(x_0))^2$	systematic error	typically falls with flexibility
$\text{Var}(\hat{f}(x_0))$	sample-to-sample instability	typically rises with flexibility

Thus, increasing flexibility typically has opposite effects:

$$\text{more flexibility} \implies \begin{cases} \text{lower bias,} \\ \text{higher variance} \end{cases}$$



We see that in every panel, squared bias (blue) decreases and variance (yellow) increases with flexibility. Their sum, plus the constant σ^2 , produces the test MSE (red) curve.



Stacking the lines, we see that:

- The grey floor σ^2 is there whatever we do
- Squared bias (blue) shrinks as we move right; variance (yellow) grows
- The optimum is where their total is the smallest (red)

The opposite dependence of bias and variance on flexibility is the **bias-variance trade-off**. At first, increasing flexibility may reduce squared bias by more than it increases variance. Eventually, the increase in variance dominates.

Low flexibility	Moderate flexibility	High flexibility
high bias	lower bias	low bias
low variance	moderate variance	high variance
underfitting	good prediction	overfitting

The optimal flexibility depends on the unknown f , the noise level, and the amount of data available.

LINEAR REGRESSION AND EXTENSIONS

2.1 Formulation and Estimation of Linear Least Squares

2.1.1 The Population Problem and Empirical Risk Minimization

The Population Problem

Let

$$X = (X_1, \dots, X_d)^\top, \quad \tilde{X} = (1, X^\top)^\top \in \mathbb{R}^p, \quad p = d + 1$$

and suppose

$$\mathbb{E}[Y^2] < \infty, \quad \mathbb{E}[\|X\|_2^2] < \infty$$

We restrict attention to linear predictors

$$g_\beta = \tilde{X}^\top \beta, \quad \beta = (\beta_0, \beta_1, \dots, \beta_d)^\top \in \mathbb{R}^p$$

and measure their prediction error by

$$L(\beta) := \mathbb{E} \left[\{Y - \tilde{X}^\top \beta\}^2 \right]$$

Definition 2.1 (Best Linear Predictor)

The best linear predictor is

$$g^*(X) = \tilde{X}^\top \beta^*, \quad \beta^* \in \arg \min_{\beta \in \mathbb{R}^p} L(\beta)$$

Optimal Conditions

Recall the loss function:

$$L(\beta) = \mathbb{E} \left[\{Y - \tilde{X}^\top \beta\}^2 \right]$$

First-order condition: Taking the gradient with respect to β gives:

$$\nabla L(\beta) = -2 \mathbb{E} \left[\tilde{X} \{Y - \tilde{X}^\top \beta\} \right]$$

Hence, setting the gradient to zero at the optimal weights β^* , we have:

$$\mathbb{E} \left[\tilde{X} \{Y - \tilde{X}^\top \beta^*\} \right] = 0$$

Second-order condition: The Hessian matrix is:

$$\nabla^2 L(\beta) = 2 \mathbb{E} [\tilde{X} \tilde{X}^\top]$$

and, for every vector $a \in \mathbb{R}^p$:

$$a^\top \nabla^2 L(\beta) a = 2 \mathbb{E} [(a^\top \tilde{X})^2] \geq 0$$

Thus, L is convex. Hence, every solution β^* is a global minimizer.

Solving the First-Order Conditions & The Solution

We begin by splitting the optimal weights vector into the intercept β_0^* and the slope coefficients $\beta_{-0}^* \in \mathbb{R}^d$:

$$\beta^* = (\beta_0^*, \beta_{-0}^{*\top})^\top$$

Expanding the first-order condition from the previous step yields two separate equations:

$$\begin{aligned} \mathbb{E}[Y - \beta_0^* - \beta_{-0}^{*\top} X] &= 0 \\ \mathbb{E}[X \{Y - \beta_0^* - \beta_{-0}^{*\top} X\}] &= 0 \end{aligned}$$

The first equation allows us to easily isolate the intercept:

$$\beta_0^* = \mathbb{E}[Y] - \beta_{-0}^{*\top} \mathbb{E}[X]$$

We then substitute this intercept back into the second equation:

$$0 = \mathbb{E}[X \{Y - \mathbb{E}[Y]\}] - \mathbb{E}[X(X - \mathbb{E}[X])^\top] \beta_{-0}^*$$

Notice that the two expectations in this equation are exactly the population covariance quantities. Because $\mathbb{E}[Y - \mathbb{E}[Y]] = 0$ and $\mathbb{E}[X - \mathbb{E}[X]] = 0$, we can rewrite them as:

$$\begin{aligned} \mathbb{E}[X \{Y - \mathbb{E}[Y]\}] &= \mathbb{E}[(X - \mathbb{E}[X])\{Y - \mathbb{E}[Y]\}] = \text{Cov}(X, Y) \\ \mathbb{E}[X(X - \mathbb{E}[X])^\top] &= \mathbb{E}[(X - \mathbb{E}[X])(X - \mathbb{E}[X])^\top] = \text{Cov}(X) \end{aligned}$$

Define $\Sigma_{XX} := \text{Cov}(X)$ and $\Sigma_{XY} := \text{Cov}(X, Y)$. The slope coefficients must therefore satisfy:

$$\Sigma_{XX} \beta_{-0}^* = \Sigma_{XY}$$

Thus, if the covariance matrix Σ_{XX} is invertible, we arrive at the final unique solution for the best linear predictor:

$$\beta_{-0}^* = \Sigma_{XX}^{-1} \Sigma_{XY}, \quad \beta_0^* = \mathbb{E}[Y] - \beta_{-0}^{*\top} \mathbb{E}[X]$$

First-Order Conditions

Define the population residual

$$\varepsilon^* := Y - g^*(X)$$

The first-order conditions become

$$\begin{aligned} \mathbb{E}[\varepsilon^*] &= 0, \\ \mathbb{E}[X_j \varepsilon^*] &= 0, \quad j = 1, \dots, d. \end{aligned}$$

Therefore,

$$\begin{aligned} \text{Cov}(X_j, \varepsilon^*) &= \mathbb{E}[X_j \varepsilon^*] - \mathbb{E}[X_j] \mathbb{E}[\varepsilon^*] \\ &= 0, \quad j = 1, \dots, d. \end{aligned}$$

The residual from the best linear predictor is **uncorrelated** with every predictor.

Non-linear Regression Function

Under the regression model,

$$f(X) = \mathbb{E}[Y | X]$$

is the best predictor among all functions of X , whereas

$$g^*(X) = \beta_0^* + \beta_{-0}^{*\top} X$$

is the best predictor among linear functions. If f is not linear, these generally differ $g^*(X) \neq f(X)$. From Theorem 1.13, the approximation error $\mathbb{E}[\{f(X) - g^*(X)\}^2]$ is 0 when $f(X) = g^*(X)$ almost surely. Restricting the model class to linear functions can create approximation error.

From the Population to the Sample

Theorem 2.2 (Weak Law of Large Numbers)

If $W_1, \dots, W_n$ are i.i.d. with

$$\mathbb{E}[W_i] = \mu, \quad \text{Var}(W_i) < \infty,$$

then, for every $\epsilon > 0$,

$$\mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n W_i - \mu\right| > \epsilon\right) \rightarrow 0.$$

In other words, as the sample size grows, the sample average converges to the population average. For example,

$$\frac{1}{n} \sum_{i=1}^n x_i y_i \approx \mathbb{E}[XY], \quad \frac{1}{n} \sum_{i=1}^n x_i x_i^\top \approx \mathbb{E}[XX^\top]$$

If the population distribution were known, we would minimise

$$L(\beta_0, \beta) = \mathbb{E} \left[\{Y - \beta_0 - \beta^\top X\}^2 \right]$$

But with i.i.d. observations $(x_i, y_i)_{i=1}^n$, replace the population average by its sample analogue:

$$L_n(\beta_0, \beta) = \frac{1}{n} \sum_{i=1}^n \{y_i - \beta_0 - \beta^\top x_i\}^2$$

Then solve the sample (a.k.a. empirical) optimisation problem: **empirical risk minimisation**

$$(\hat{\beta}_0, \hat{\beta}) \in \arg \min_{\beta_0, \beta} L_n(\beta_0, \beta)$$

Solving the Sample Problem: One Predictor

For a single predictor ($d = 1$), the empirical loss function is:

$$L_n(\beta_0, \beta_1) = \frac{1}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

First-order condition: Setting the partial derivatives to zero:

$$\begin{aligned} \frac{\partial L_n}{\partial \beta_0} &= -\frac{2}{n} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0 \\ \frac{\partial L_n}{\partial \beta_1} &= -\frac{2}{n} \sum_{i=1}^n x_i (y_i - \beta_0 - \beta_1 x_i) = 0 \end{aligned}$$

Thus, any minimizer $(\hat{\beta}_0, \hat{\beta}_1)$ must satisfy:

$$\begin{aligned} \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) &= 0 \\ \sum_{i=1}^n x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) &= 0 \end{aligned}$$

Second-order condition: The Hessian matrix is:

$$\nabla^2 L_n(\beta_0, \beta_1) = \frac{2}{n} \begin{pmatrix} n & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \end{pmatrix}$$

For any vector $(a, b) \in \mathbb{R}^2$:

$$(a, b) \nabla^2 L_n(\beta_0, \beta_1) \begin{pmatrix} a \\ b \end{pmatrix} = \frac{2}{n} \sum_{i=1}^n (a + bx_i)^2 \geq 0$$

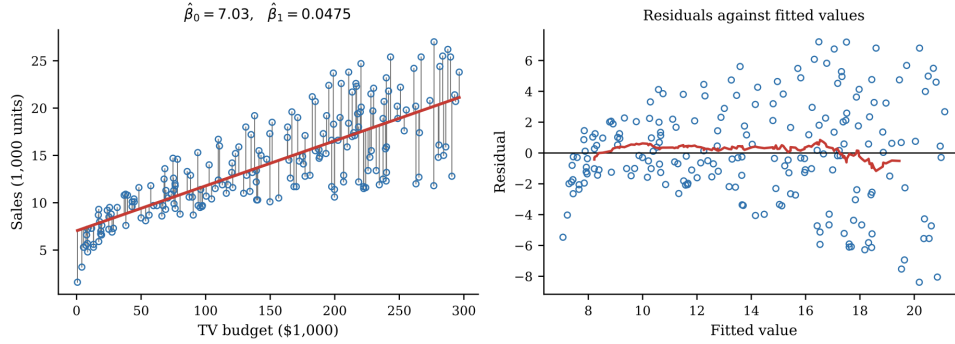
Hence, L_n is convex, and $(\hat{\beta}_0, \hat{\beta}_1)$ is a global minimizer.

Solving the equations: The first normal equation gives:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}, \quad \text{where } \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

Substituting this into the second equation:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$



- **Left:** the least squares line minimised the total squared length of the grey segments/
- **Right:** the residuals $e_i = y_i - \hat{y}_i$ against the fitted values $\hat{y}_i$. By construction, they average to zero and are orthogonal to the fitted values: $\sum_{i=1}^n e_i = 0 \Rightarrow \bar{e} = 0$

2.1.2 Matrix Foundations and Least Squares Solution

Matrix Toolbox

Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$.

Definition 2.3 (Inner Product)

The **Euclidean inner product** of $\mathbf{u}$ and $\mathbf{v}$ is

$$\langle \mathbf{u}, \mathbf{v} \rangle := \mathbf{u}^\top \mathbf{v} = \sum_{i=1}^n u_i v_i$$

Definition 2.4 (Euclidean Norm)

The **Euclidean norm** of $\mathbf{v}$ is

$$\|\mathbf{v}\|_2 := \sqrt{\mathbf{v}^\top \mathbf{v}} = \left(\sum_{i=1}^n v_i^2 \right)^{1/2}$$

Hence,

$$\|\mathbf{v}\|_2^2 = \mathbf{v}^\top \mathbf{v}$$

(The Euclidean norm measures the length of a vector.)

Let $\mathbf{A} \in \mathbb{R}^{n \times p}$, with columns $\mathbf{a}_1, \dots, \mathbf{a}_p \in \mathbb{R}^n$.

Definition 2.5 (Column Space and Rank)

The **column space** and **rank** of $\mathbf{A}$ are

$$\text{col}(\mathbf{A}) = \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_p\}, \quad \text{rank}(\mathbf{A}) = \dim\{\text{col}(\mathbf{A})\}.$$

Thus, $\text{rank}(\mathbf{A}) = p$ exactly when the columns of $\mathbf{A}$ are linearly independent.

Definition 2.6 (Orthogonality) Two vectors are **orthogonal** if

$$\mathbf{u} \perp \mathbf{v} \iff \mathbf{u}^\top \mathbf{v} = 0$$

Moreover,

$$\mathbf{v} \perp \text{col}(\mathbf{A}) \iff \mathbf{A}^\top \mathbf{v} = \mathbf{0} \iff \mathbf{a}_j^\top \mathbf{v} = 0, \quad j = 1, \dots, p.$$

Let $\mathbf{A} \in \mathbb{R}^{p \times p}$ be symmetric.

Property	Notation	Definition
positive definite	$\mathbf{A} \succ 0$	$\mathbf{v}^\top \mathbf{A} \mathbf{v} > 0$ for all $\mathbf{v} \neq \mathbf{0}$
positive semidefinite	$\mathbf{A} \succeq 0$	$\mathbf{v}^\top \mathbf{A} \mathbf{v} \geq 0$ for all $\mathbf{v}$
negative definite	$\mathbf{A} \prec 0$	$\mathbf{v}^\top \mathbf{A} \mathbf{v} < 0$ for all $\mathbf{v} \neq \mathbf{0}$
negative semidefinite	$\mathbf{A} \preceq 0$	$\mathbf{v}^\top \mathbf{A} \mathbf{v} \leq 0$ for all $\mathbf{v}$
indefinite		$\mathbf{v}^\top \mathbf{A} \mathbf{v}$ takes both signs

For every $\mathbf{X} \in \mathbb{R}^{n \times p}$,

$$\mathbf{v}^\top \mathbf{X}^\top \mathbf{X} \mathbf{v} = (\mathbf{X} \mathbf{v})^\top (\mathbf{X} \mathbf{v}) = \|\mathbf{X} \mathbf{v}\|_2^2 \geq 0$$

so

$$\mathbf{X}^\top \mathbf{X} \succeq 0.$$

Moreover,

$$\mathbf{X}^\top \mathbf{X} \succ 0 \iff \text{rank}(\mathbf{X}) = p \iff \mathbf{X}^\top \mathbf{X} \text{ is invertible.}$$

Definition 2.7 (Trace)

Let $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{p \times p}$. The **trace** of $\mathbf{A}$ is the sum of its diagonal entries:

$$\text{tr}(\mathbf{A}) := \sum_{j=1}^p A_{jj}.$$

The trace is linear: $\text{tr}(a\mathbf{A} + b\mathbf{B}) = a \text{tr}(\mathbf{A}) + b \text{tr}(\mathbf{B})$ for all $a, b \in \mathbb{R}$.

Theorem 2.8 (Cyclicity of the Trace)

Let $\mathbf{A} \in \mathbb{R}^{p \times q}$, $\mathbf{B} \in \mathbb{R}^{q \times r}$, $\mathbf{C} \in \mathbb{R}^{r \times p}$. Then,

$$\text{tr}(\mathbf{ABC}) = \text{tr}(\mathbf{BCA}) = \text{tr}(\mathbf{CAB}).$$

More generally, factors inside a trace may be cyclically permuted whenever their products are well defined.

Trivial (but important) fact: If $\mathbf{A} \in \mathbb{R}^{1 \times 1}$ (i.e. $\mathbf{A}$ is a scalar), then $\text{tr}(\mathbf{A}) = \mathbf{A}$.

Matrix Notation

For observation i , write

$$X_i = (x_{i1}, \dots, x_{id})^\top \in \mathbb{R}^d$$

for its observed predictor vector. Collect the n observations in the design matrix

$$X = \begin{pmatrix} 1 & X_1^\top \\ 1 & X_2^\top \\ \vdots & \vdots \\ 1 & X_n^\top \end{pmatrix} = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1d} \\ 1 & x_{21} & \cdots & x_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{nd} \end{pmatrix} \in \mathbb{R}^{n \times p}, \quad p = d + 1.$$

Define

$$y = (y_1, \dots, y_n)^\top, \quad \beta = (\beta_0, \beta_1, \dots, \beta_d)^\top.$$

Then

$$y = X\beta + \epsilon, \quad L_n(\beta) := \frac{1}{n} \|y - X\beta\|_2^2.$$

Solving the Least-Squares Problem

The least-squares objective function in matrix notation is:

$$L_n(\beta) = \frac{1}{n} \|y - X\beta\|_2^2 = \frac{1}{n} (y - X\beta)^\top (y - X\beta)$$

First-order condition: Setting the gradient with respect to β to zero:

$$\nabla L_n(\beta) = -\frac{2}{n} X^\top (y - X\beta) = \mathbf{0}$$

This yields the **normal equations**:

$$X^\top X \hat{\beta} = X^\top y$$

Second-order condition: The Hessian matrix is:

$$\nabla^2 L_n(\beta) = \frac{2}{n} X^\top X$$

For any non-zero vector $v \in \mathbb{R}^p$:

$$v^\top \nabla^2 L_n(\beta) v = \frac{2}{n} \|Xv\|_2^2 \geq 0$$

Thus, L_n is convex. If $\text{rank}(X) = p$, then $X^\top X$ is invertible and the minimiser is unique:

$$\hat{\beta} = (X^\top X)^{-1} X^\top y$$

Once $\hat{\beta}$ has been computed, the vector of fitted values is

$$\hat{\mathbf{y}} := (\hat{y}_1, \dots, \hat{y}_n)^\top = X\hat{\beta} = X(X^\top X)^{-1} X^\top y$$

Definition 2.9 (Hat Matrix)

The matrix

$$H := X(X^\top X)^{-1} X^\top$$

is called the hat matrix.

Thus, the fitted values and residuals are

$$\hat{\mathbf{y}} = H\mathbf{y}, \quad \mathbf{e} := \mathbf{y} - \hat{\mathbf{y}} = (I - H)\mathbf{y}$$

where H maps the observed responses $\mathbf{y}$ to their fitted values $\hat{\mathbf{y}}$.

2.2 Geometry, Projection, and Error Analysis

2.2.1 Orthogonality and Geometric Projection

Design Matrix

Write the columns of the design matrix as

$$X = \begin{pmatrix} 1 & X_1^\top \\ \vdots & \vdots \\ 1 & X_n^\top \end{pmatrix} = (1, X_1, \dots, X_d),$$

where

$$X_j = (x_{1j}, \dots, x_{nj})^\top \in \mathbb{R}^n$$

is the vector containing the observed values of predictor j , and

$$\mathbf{1} = (1, \dots, 1)^\top \in \mathbb{R}^n,$$

is the intercept vector of one's.

Every fitted response vector has the form

$$X\beta = \beta_0 \mathbf{1} + \beta_1 X_1 + \dots + \beta_d X_d$$

Hence, all fitted vectors lie in

$$\text{col}(X) = \text{span}\{\mathbf{1}, X_1, \dots, X_d\} \subset \mathbb{R}^n.$$

If $\text{rank}(X) = p = d + 1$, this is a p -dimensional subspace of $\mathbb{R}^n$.

There are two useful ways to read the same matrix $\mathbf{X}$. The *Rows are the Observations*: The i th row is $(1, X_i^\top) = (1, x_{i1}, \dots, x_{id})$, where $X_i \in \mathbb{R}^d$ contains all predictors for observation i . The *Columns are the Variables*: The j th predictor column is $\mathbf{X}_j = (x_{1j}, \dots, x_{nj})^\top \in \mathbb{R}^n$ which contains predictor j across all observations. The first column $\mathbf{1}$ is the intercept.

Orthogonality of the Residual

The sample first-order condition (the normal equations) states

$$X^\top (y - X\hat{\beta}) = 0$$

Since $\hat{\mathbf{y}} = X\hat{\beta}$ and $\mathbf{e} := \mathbf{y} - \hat{\mathbf{y}}$, this becomes

$$X^\top \mathbf{e} = 0$$

Writing $X = (1, X_1, \dots, X_d)$, this is equivalent to

$$\mathbf{1}^\top \mathbf{e} = 0, \quad X_j^\top \mathbf{e} = 0, \quad j = 1, \dots, d$$

Thus, $\mathbf{e}$ is orthogonal to every column of X , and therefore to every linear combination of its columns:

$$\mathbf{e} \perp \text{col}(X)$$

The least-squares residual is orthogonal to the entire space of fitted response vectors.

Least Squares as a Geometric Problem

Least squares minimises

$$\min_{\beta \in \mathbb{R}^p} \|y - X\beta\|_2^2$$

Since $X\beta$ ranges over $\text{col}(X)$, this is equivalent to

$$\min_{v \in \text{col}(X)} \|y - v\|_2^2$$

So the algebraic problem has a geometric interpretation: *which point in $\text{col}(X)$ is closest to y ?* The closest point is called the **orthogonal projection** of y onto $\text{col}(X)$. For least squares, this point is $\hat{y}$, since

$$\hat{y} \in \text{col}(X), \quad y - \hat{y} = e \perp \text{col}(X)$$

$\hat{y}$ lies in the fitted space, and the remaining error is orthogonal to that space.

$\hat{y} = Hy \in \text{col}(X)$, and $e = y - \hat{y} \perp \text{col}(X)$. The equation $X^\top e = 0$ is exactly this orthogonality condition.

Theorem 2.10 (Projection Characterization of Least Squares)

A vector $v \in \text{col}(X)$ minimises $\|y - u\|^2$ over $u \in \text{col}(X)$ if and only if

$$y - v \perp \text{col}(X)$$

If $\text{rank}(X) = p$, the minimiser is $v = \hat{y} = Hy$.

Properties of the Hat Matrix

Proposition 2.11

If $\text{rank}(X) = p$, then:

1. **Symmetry:** $H^\top = H$
2. **Idempotency:** $H^2 = H$
3. **Identity on the fitted space:** $HX = X$
4. **Orthogonality of fitted values and residuals:** $H(I - H) = 0$
5. **Rank/trace:** $\text{tr}(H) = p$

Proof.

1. **Symmetry.** Since $X^\top X$ is symmetric, so is $(X^\top X)^{-1}$. Hence

$$H^\top = [X(X^\top X)^{-1}X^\top]^\top = X[(X^\top X)^{-1}]^\top X^\top = X(X^\top X)^{-1}X^\top = H$$

2. **Idempotency.** Directly,

$$\begin{aligned} H^2 &= X(X^\top X)^{-1}X^\top \cdot X(X^\top X)^{-1}X^\top \\ &= X(X^\top X)^{-1}(X^\top X)(X^\top X)^{-1}X^\top \\ &= X(X^\top X)^{-1}X^\top = H \end{aligned}$$

3. **Identity on the fitted space.** Directly,

$$HX = X(X^\top X)^{-1}X^\top X = X(X^\top X)^{-1}(X^\top X) = X$$

4. **Orthogonality of fitted values and residuals.** Using idempotency,

$$H(I - H) = H - H^2 = H - H = 0$$

Equivalently, for any y , $(Hy)^\top (I - H)y = y^\top H^\top (I - H)y = y^\top H(I - H)y = 0$, using symmetry of H . So $\hat{y} = Hy$ and $e = (I - H)y$ are orthogonal, consistent with $e \perp \text{col}(X)$.

5. **Rank/trace.** By the cyclicity of the trace (Theorem ??),

$$\text{tr}(H) = \text{tr}(X(X^\top X)^{-1}X^\top) = \text{tr}((X^\top X)^{-1}X^\top X) = \text{tr}(I_p) = p$$

Since H is idempotent, its eigenvalues are only 0 or 1, so $\text{rank}(H) = \text{tr}(H) = p$.

□

Decomposing Sample Variation: The Pythagorean Identity and R^2

Recall that the law of total variance:

$$\text{Var}(Y) = \underbrace{\text{Var}(\mathbb{E}[Y | X])}_{\text{explained by } X} + \underbrace{\mathbb{E}[\text{Var}(Y | X)]}_{\text{remaining given } X}$$

Under the homoscedastic regression model, $\mathbb{E}[Y | X] = f(X)$ and $\text{Var}(Y | X) = \sigma^2$, so

$$\text{Var}(Y) = \text{Var}(f(X)) + \sigma^2$$

Least squares gives a closely related **sample** decomposition. Since the model contains an intercept, $\bar{\hat{y}} = \bar{y}$, and

$$y - \bar{y}\mathbf{1} = (\hat{y} - \bar{y}\mathbf{1}) + e, \quad (\hat{y} - \bar{y}\mathbf{1}) \perp e$$

Hence, by Pythagoras,

$$\underbrace{\sum_{i=1}^n (y_i - \bar{y})^2}_{\text{total}} = \underbrace{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}_{\text{explained by the fitted model}} + \underbrace{\sum_{i=1}^n (y_i - \hat{y}_i)^2}_{\text{remaining}}$$

Write

$$\text{TSS} := \sum_{i=1}^n (y_i - \bar{y})^2, \quad \text{ESS} := \sum_{i=1}^n (\hat{y}_i - \bar{y})^2, \quad \text{RSS} := \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Then

$$\text{TSS} = \text{ESS} + \text{RSS}$$

and therefore

$$R^2 := \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}$$

Thus, R^2 is the fraction of the observed sample variation in Y explained by the fitted linear model. R^2 measures in-sample fit; it is **not** a measure of out-of-sample prediction accuracy.

2.2.2 Bias and Variance

Training and Test Error

Let

$$y_{0,i} = f(X_i) + \epsilon_{0,i}, \quad i = 1, \dots, n,$$

be new responses at the same design points $X_1, \dots, X_n$, independent of the sample responses $y_1, \dots, y_n$. In vector form,

$$y_0 = f + \varepsilon_0$$

Define the training and in-sample prediction errors by

$$\text{MSE}_{\text{train}} := \frac{1}{n} \|y - \hat{f}\|_2^2, \quad \text{MSE}_{\text{test}} := \frac{1}{n} \|y_0 - \hat{f}\|_2^2, \quad \hat{f} = Hy$$

$$\mathbb{E}[\text{MSE}_{\text{train}}] = \frac{1}{n} \|(I - H)f\|_2^2 + \sigma^2 \left(1 - \frac{p}{n}\right),$$

$$\mathbb{E}[\text{MSE}_{\text{test}}] = \frac{1}{n} \|(I - H)f\|_2^2 + \sigma^2 \left(1 + \frac{p}{n}\right).$$

Expected Training Error: Derivation

Since $y = f + \varepsilon$ and $\hat{f} = Hy$,

$$y - \hat{f} = (I - H)f + (I - H)\varepsilon$$

Hence,

$$\begin{aligned} \|y - \hat{f}\|_2^2 &= \|(I - H)f\|_2^2 + \|(I - H)\varepsilon\|_2^2 + 2f^\top (I - H)^\top (I - H)\varepsilon \\ &= \|(I - H)f\|_2^2 + \|(I - H)\varepsilon\|_2^2 + 2f^\top (I - H)\varepsilon, \end{aligned}$$

where the last equality uses symmetry and idempotency of H . Using $E[\varepsilon] = 0$, the cross term has mean zero. Therefore,

$$E\|y - \hat{f}\|_2^2 = \|(I - H)f\|_2^2 + E\|(I - H)\varepsilon\|_2^2$$

It remains to calculate the second term:

$$\begin{aligned} E\|(I - H)\varepsilon\|_2^2 &= E[\varepsilon^\top (I - H)^\top (I - H)\varepsilon] \\ &\stackrel{(a)}{=} E[\varepsilon^\top (I - H)\varepsilon] \\ &= E[\text{tr}\{(I - H)\varepsilon\varepsilon^\top\}] \\ &= \text{tr}\{(I - H) E[\varepsilon\varepsilon^\top]\} \\ &\stackrel{(b)}{=} \sigma^2 \text{tr}(I - H) \\ &= \sigma^2 \{n - \text{tr}(H)\} \\ &\stackrel{(c)}{=} \sigma^2(n - p), \end{aligned}$$

where (a) holds by symmetry and idempotency of H , (b) because $E[\varepsilon\varepsilon^\top] = \sigma^2 I$, and (c) since $\text{tr}(H) = \text{rank}(H) = p$.

Therefore,

$$E\|y - \hat{f}\|_2^2 = \|(I - H)f\|_2^2 + \sigma^2(n - p),$$

and hence

$$E[\text{MSE}_{\text{train}}] = \frac{1}{n} \|(I - H)f\|_2^2 + \sigma^2 \left(1 - \frac{p}{n}\right)$$

Expected Test Error: Derivation

Recall $E[\varepsilon_0\varepsilon_0^\top] = E[\varepsilon\varepsilon^\top] = \sigma^2 I$. For the new-response noise,

$$E\|\varepsilon_0\|_2^2 = E[\text{tr}(\varepsilon_0\varepsilon_0^\top)] = \text{tr}\{E[\varepsilon_0\varepsilon_0^\top]\} = \sigma^2 \text{tr}(I) = n\sigma^2$$

For the estimation noise, using cyclicity of the trace,

$$E\|H\varepsilon\|_2^2 = E[\text{tr}\{H^\top H\varepsilon\varepsilon^\top\}] = \text{tr}\{H^\top H E[\varepsilon\varepsilon^\top]\} = \sigma^2 \text{tr}(H^2) = \sigma^2 \text{tr}(H) = \sigma^2 p,$$

where $H^\top = H$, $H^2 = H$, and $\text{tr}(H) = p$. Putting everything together,

$$E\|y_0 - \hat{f}\|_2^2 = \underbrace{\|(I - H)f\|_2^2}_{\text{bias}^2} + \underbrace{n\sigma^2}_{\text{new-response noise}} + \underbrace{\sigma^2 p}_{\text{estimation variance}}$$

Thus,

$$E[\text{MSE}_{\text{test}}] = \sigma^2 + \frac{1}{n} \|(I - H)f\|_2^2 + \frac{\sigma^2 p}{n}$$

Bias-Variance Trade-off for Linear Least Squares

$$\mathbb{E}[\text{MSE}_{\text{test}}] = \underbrace{\sigma^2}_{\text{irreducible}} + \underbrace{\frac{1}{n} \|(I - H)f\|_2^2}_{\text{squared bias}} + \underbrace{\frac{\sigma^2 p}{n}}_{\text{variance}}$$

Suppose we enlarge the linear model by adding one predictor, so that $\text{col}(X_p) \subset \text{col}(X_{p+1})$, and let H_p and H_{p+1} denote the associated hat matrices. Since least squares projects f onto the corresponding column spaces,

$$\|(I - H_{p+1})f\|_2^2 \leq \|(I - H_p)f\|_2^2$$

Thus, increasing model flexibility can only decrease the squared bias. At the same time, the variance increases by

$$\frac{\sigma^2(p+1)}{n} - \frac{\sigma^2 p}{n} = \frac{\sigma^2}{n}$$

More flexibility $\implies$ bias² $\downarrow$ but variance $\uparrow$. The U-shape arises from the competition between these two terms.

Why Does Flexibility Reduce Squared Bias?

Suppose that $\text{col}(X_p) \subseteq \text{col}(X_{p+1})$, and let H_p and H_{p+1} be the corresponding orthogonal projection matrices. Then

$$(I - H_p)f = (I - H_{p+1})f + (H_{p+1} - H_p)f$$

The two terms on the right are orthogonal, so

$$\|(I - H_p)f\|_2^2 = \|(I - H_{p+1})f\|_2^2 + \|(H_{p+1} - H_p)f\|_2^2$$

Therefore,

$$\|(I - H_{p+1})f\|_2^2 \leq \|(I - H_p)f\|_2^2$$

Enlarging the model cannot increase its in-sample squared bias. The decrease is strict exactly when

$$(H_{p+1} - H_p)f \neq 0,$$

i.e. when the new direction captures part of f not already contained in $\text{col}(X_p)$.

2.3 Statistical Inference and Hypothesis Testing

2.3.1 Sampling Variability and Confidence Intervals

Variance of the OLS Estimator

Lemma 2.12 (Mean and Variance under Linear Transformations)

Let $y \in \mathbb{R}^n$ be a random vector with finite second moments, and let $A \in \mathbb{R}^{m \times n}$ be fixed. Then

$$\mathbb{E}[Ay] = A \mathbb{E}[y], \quad \text{Var}(Ay) = A \text{Var}(y) A^\top$$

Proof.

By linearity of expectation,

$$\mathbb{E}[Ay] = A \mathbb{E}[y]$$

Moreover,

$$Ay - \mathbb{E}[Ay] = A\{y - \mathbb{E}[y]\}$$

Hence,

$$\begin{aligned} \text{Var}(Ay) &= \mathbb{E}[\{Ay - \mathbb{E}[Ay]\}\{Ay - \mathbb{E}[Ay]\}^\top] \\ &= \mathbb{E}[A\{y - \mathbb{E}[y]\}\{y - \mathbb{E}[y]\}^\top A^\top] \\ &= A \mathbb{E}[\{y - \mathbb{E}[y]\}\{y - \mathbb{E}[y]\}^\top] A^\top \\ &= A \text{Var}(y) A^\top \end{aligned}$$

□

Definition 2.13 (Unbiased Estimators)

An estimator $\hat{\theta}$ of a parameter θ is **unbiased** if

$$E[\hat{\theta}] = \theta$$

Its bias is

$$\text{Bias}(\hat{\theta}) := E[\hat{\theta}] - \theta$$

Thus, $\hat{\theta}$ is unbiased if and only if $\text{Bias}(\hat{\theta}) = 0$

Over repeated samples, an unbiased estimator is **centered at the true parameter**: it has no systematic tendency to overestimate or underestimate θ . Unbiased does not necessarily mean best: recall the bias-variance decomposition $\text{MSE}(\hat{\theta}) = \text{Bias}(\hat{\theta})^2 + \text{Var}(\hat{\theta})$, an estimator with some bias may have substantially smaller variance.

Consider the linear model

$$\mathbf{y} = X\boldsymbol{\beta} + \varepsilon, \quad E[\varepsilon] = 0, \quad \text{Var}(\varepsilon) = \sigma^2 I_n$$

where $X \in \mathbb{R}^{n \times p}$ is fixed and $\text{rank}(X) = p$.

Theorem 2.14 (First and Second Moments of OLS)

The least-squares estimator $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}$ satisfies

$$E[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}, \quad \text{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2 (X^\top X)^{-1}$$

The matrix $\text{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2 (X^\top X)^{-1}$ answers several questions at once.

1. **Diagonal Entries:** If $v_{jj} = [(X^\top X)^{-1}]_{jj}$ then $\text{Var}(\hat{\beta}_j) = \sigma^2 v_{jj}$.
Every usual coefficient standard error estimate $\sigma \sqrt{v_{jj}}$.
2. **Off-diagonal Entries:** $\text{Cov}(\hat{\beta}_j, \hat{\beta}_k) = \sigma^2 [(X^\top X)^{-1}]_{jk}$
The coefficient estimates are generally correlated; they are uncorrelated when the relevant columns of X are orthogonal.

The noise level enters through σ^2 , while $(X^\top X)^{-1}$ reflects the observed predictor values. More variation in the predictors generally makes their effects easier to distinguish and estimate precisely; little variation makes estimation less precise. This leaves σ^2 as the only unknown component.

Consider the linear model

$$\mathbf{y} = X\boldsymbol{\beta} + \varepsilon, \quad E[\varepsilon] = 0, \quad \text{Var}(\varepsilon) = \sigma^2 I_n$$

where $X \in \mathbb{R}^{n \times p}$ is fixed and $\text{rank}(X) = p$.

Theorem 2.15 (Unbiased estimator of σ^2)

Under the standing assumptions,

$$s^2 := \frac{RSS}{n - p} = \frac{\|(I - H)\mathbf{y}\|_2^2}{n - p}$$

satisfies

$$E[s^2] = \sigma^2$$

Its square root is called the **residual standard error**.

Proof.

Since $(I - H)X = 0$, the residual vector satisfies

$$\mathbf{e} = (I - H)\mathbf{y} = (I - H)(X\boldsymbol{\beta} + \varepsilon) = (I - H)\varepsilon$$

Therefore,

$$RSS = \|\mathbf{e}\|_2^2 = \|(I - H)\varepsilon\|_2^2$$

Using the same calculation as before,

$$E[\text{RSS}] = \sigma^2 \text{tr}(I - H) = \sigma^2 \{n - \text{tr}(H)\} = \sigma^2(n - p),$$

since $\text{tr}(H) = p$. Hence

$$E[s^2] = \frac{E[\text{RSS}]}{n - p} = \sigma^2$$

□

The Gauss-Markov Theorem

Consider the linear model

$$\mathbf{y} = X\boldsymbol{\beta} + \varepsilon, \quad E[\varepsilon] = 0, \quad \text{Var}(\varepsilon) = \sigma^2 I_n$$

where $X \in \mathbb{R}^{n \times p}$ is fixed and $\text{rank}(X) = p$.

Theorem 2.16 (Gauss-Markov Theorem)

Let $c \in \mathbb{R}^p$. Among all **linear unbiased estimators**

$$a^\top \mathbf{y} = \sum_{i=1}^n a_i Y_i, \quad a \in \mathbb{R}^n$$

of $c^\top \boldsymbol{\beta}$, the least-squares estimator $c^\top \hat{\boldsymbol{\beta}}$ has the smallest variance.

Recall that $\hat{\theta}$ is an unbiased estimate for θ if $E[\hat{\theta}] = \theta$. Taking $c = e_j$, the j th standard basis vector, shows that no linear unbiased estimator of β_j has smaller variance than $\hat{\beta}_j$. *Least squares is the Best Linear Unbiased Estimator (BLUE)*.

Proof.

Let $a^\top \mathbf{y}$ be any linear unbiased estimator of $c^\top \boldsymbol{\beta}$. Since

$$E[a^\top \mathbf{y}] = a^\top X\boldsymbol{\beta},$$

unbiasedness for every $\boldsymbol{\beta} \in \mathbb{R}^p$ requires

$$(X^\top a - c)^\top \boldsymbol{\beta} = 0 \quad \text{for every } \boldsymbol{\beta}$$

If $X^\top a - c \neq 0$, choosing $\boldsymbol{\beta} = X^\top a - c$ gives

$$0 = \|X^\top a - c\|_2^2 > 0,$$

a contradiction. Hence, $X^\top a = c$.

The OLS estimator can be written as

$$c^\top \hat{\boldsymbol{\beta}} = a_{\text{OLS}}^\top \mathbf{y}, \quad a_{\text{OLS}} := X(X^\top X)^{-1}c$$

For any competing a , write $d := a - a_{\text{OLS}}$. Then,

$$X^\top d = X^\top a - c = 0$$

Using the definition of a_{OLS} ,

$$a_{\text{OLS}}^\top d = c^\top (X^\top X)^{-1} X^\top d = 0$$

Therefore,

$$\|a\|_2^2 = \|a_{\text{OLS}} + d\|_2^2 = \|a_{\text{OLS}}\|_2^2 + \|d\|_2^2$$

Finally, since $\text{Var}(y) = \sigma^2 I_n$,

$$\begin{aligned} \text{Var}(a^\top y) &= \sigma^2 \|a\|_2^2 \\ &= \sigma^2 \|a_{\text{OLS}}\|_2^2 + \sigma^2 \|d\|_2^2 \\ &\geq \sigma^2 \|a_{\text{OLS}}\|_2^2 \\ &= \text{Var}(a_{\text{OLS}}^\top y) \\ &= \text{Var}(c^\top \hat{\beta}) \end{aligned}$$

□

Each of the three qualifiers is doing work. **Best** refers to variance only. Among unbiased estimators this is the same as mean squared error, but no further. **Linear** means linear in $\mathbf{y}$. Non-linear estimators are not covered. **Unbiased** is a genuine restriction and one worth questioning. Recall that $\text{MSE} = \text{Bias}^2 + \text{Var}$. Gauss-Markov fixes the first term at zero and minimises the second. Nothing says this minimises the sum. Accepting a little bias can buy a large reduction in variance.

Maximum Likelihood under Gaussian Errors

Definition 2.17 (Maximum Likelihood Principle)

Given a dataset, choose the parameter(s) of interest in such a way that the data are most likely.

Definition 2.18 (Likelihood Function)

Suppose that the random sample $X_1, \dots, X_n$ has joint pdf or pmf

$$f(x_1, \dots, x_n; \theta_1, \dots, \theta_m) \equiv f(\mathbf{x}; \boldsymbol{\theta})$$

Define the **likelihood function** as

$$L(\boldsymbol{\theta}) := L(\theta_1, \dots, \theta_m) = f(x_1, \dots, x_n; \theta_1, \dots, \theta_m)$$

This function shows how likely it is to observe the data set $x_1, \dots, x_n$. For a random sample from pdf (or pmf) $f(x; \boldsymbol{\theta})$,

$$L(\boldsymbol{\theta}) = \prod_{i=1}^n f(x_i; \boldsymbol{\theta}) = f(x_1; \boldsymbol{\theta}) f(x_2; \boldsymbol{\theta}) \cdots f(x_n; \boldsymbol{\theta})$$

Density and likelihood are two sides of the same coin: density regards $f(x; \boldsymbol{\theta})$ as a function of x with known $\boldsymbol{\theta}$ (probability); likelihood regards $f(x; \boldsymbol{\theta})$ as a function of $\boldsymbol{\theta}$ while the data x is given (statistics).

Definition 2.19 (Maximum Likelihood Estimator)

Let $x_1, \dots, x_n$ be a realization of a random sample with joint pdf or pmf $f(\cdot; \boldsymbol{\theta})$. The **maximum likelihood estimate (MLE)** of $\boldsymbol{\theta} \in \Theta$ is the maximizer $\hat{\boldsymbol{\theta}}$ of the likelihood function $L(\boldsymbol{\theta})$, i.e.

$$\hat{\boldsymbol{\theta}} = \arg \max_{\boldsymbol{\theta} \in \Theta} L(\boldsymbol{\theta})$$

The same notation denotes the likelihood function evaluated at a realized data set $x_1, \dots, x_n$ and at a random sample $X_1, \dots, X_n$: the **maximum likelihood estimator** is the maximizer of $L(\boldsymbol{\theta})$ evaluated at $X_1, \dots, X_n$.

The MLE $\hat{\boldsymbol{\theta}}$ is also the maximizer of the **log-likelihood**,

$$\hat{\boldsymbol{\theta}} = \arg \max_{\boldsymbol{\theta} \in \Theta} \log L(\boldsymbol{\theta})$$

Assume now, in addition, that $\varepsilon \sim N(0, \sigma^2 I_n)$, equivalently $\mathbf{y} \sim N(X\boldsymbol{\beta}, \sigma^2 I_n)$. The likelihood is

$$L(\boldsymbol{\beta}, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(y_i - X_i^\top \boldsymbol{\beta})^2}{2\sigma^2} \right\}$$

Thus

$$\ell(\boldsymbol{\beta}, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2$$

For fixed σ^2 , maximising ℓ in $\boldsymbol{\beta}$ is exactly the same as minimising squared error:

$$\hat{\boldsymbol{\beta}}_{\text{MLE}} = \hat{\boldsymbol{\beta}}_{\text{OLS}} = (X^\top X)^{-1} X^\top \mathbf{y}$$

Substituting $\hat{\boldsymbol{\beta}}$ and differentiating in σ^2 ,

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{\text{RSS}}{2\sigma^4} = 0,$$

so

$$\hat{\sigma}_{\text{MLE}}^2 = \frac{\text{RSS}}{n}$$

But $E[\text{RSS}] = \sigma^2(n-p)$, hence

$$E[\hat{\sigma}_{\text{MLE}}^2] = \frac{n-p}{n} \sigma^2 < \sigma^2$$

Thus the MLE of σ^2 is biased downward. In contrast, $s^2 = \text{RSS}/(n-p)$ is unbiased and is the variance estimate used in the usual regression standard errors.

Adding Normality

Consider the linear model

$$\mathbf{y} = X\boldsymbol{\beta} + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2 I_n),$$

where $X \in \mathbb{R}^{n \times p}$ is fixed and $\text{rank}(X) = p$. The normality assumption gives us the exact sampling distributions needed for classical statistical inference.

Result 2.3.1 (Distribution Theory for the Linear Model)

1. $\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \sigma^2(X^\top X)^{-1})$
2. $\frac{\text{RSS}}{\sigma^2} = \frac{(n-p)s^2}{\sigma^2} \sim \chi_{n-p}^2$
3. $\hat{\boldsymbol{\beta}}$ and RSS are independent.

We will use these results without proof. Their proofs are topics in theoretical statistics courses.

Confidence Intervals via the t-Pivot

Let $v_{jj} := [(X^\top X)^{-1}]_{jj}$. Then

$$\frac{\hat{\beta}_j - \beta_j}{\sigma \sqrt{v_{jj}}} \sim N(0, 1)$$

Since σ is unknown, replace it by

$$s = \sqrt{\frac{\text{RSS}}{n-p}}$$

Independence of $\hat{\boldsymbol{\beta}}$ and s^2 gives:

Result 2.3.2 (The t-Pivot)

$$\frac{\hat{\beta}_j - \beta_j}{SE(\hat{\beta}_j)} \sim t_{n-p}, \quad SE(\hat{\beta}_j) = s\sqrt{v_{jj}},$$

where t_{n-p} denotes the t-distribution with $n - p$ degrees of freedom.

This result will give us both confidence intervals and hypothesis tests.

A point estimate such as $\hat{\beta}_j$ gives one estimate of the unknown coefficient β_j , but it changes from sample to sample.

Definition 2.20 (Confidence Interval)

A $(1 - \alpha)$ confidence interval is a random interval

$$C(\mathbf{y}) = [L(\mathbf{y}), U(\mathbf{y})]$$

constructed so that

$$P\{\beta_j \in C(\mathbf{y})\} = 1 - \alpha$$

This probability refers to *repeated sampling*: if we repeatedly collected new data and constructed the interval in the same way, a proportion $1 - \alpha$ of the intervals would contain the true coefficient. Once the data has been observed, the interval is fixed and β_j is fixed: the interval either contains β_j or not.

From

$$\frac{\hat{\beta}_j - \beta_j}{SE(\hat{\beta}_j)} \sim t_{n-p},$$

we have

$$P\left(-t_{n-p, 1-\alpha/2} \leq \frac{\hat{\beta}_j - \beta_j}{SE(\hat{\beta}_j)} \leq t_{n-p, 1-\alpha/2}\right) = 1 - \alpha$$

Rearranging gives:

Result 2.3.3 (Confidence Interval for β_j)

$$\hat{\beta}_j \pm t_{n-p, 1-\alpha/2} SE(\hat{\beta}_j),$$

where $t_{n-p, 1-\alpha/2}$ is the $1 - \alpha/2$ quantile of the t-distribution with $n - p$ degrees of freedom.

At a new predictor vector $\mathbf{x}_0$, the conditional mean is

$$\mu_0 := E[Y_0 | \mathbf{x}_0] = \mathbf{x}_0^\top \boldsymbol{\beta},$$

estimated by

$$\hat{\mu}_0 = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$$

Its estimation error is

$$\hat{\mu}_0 - \mu_0 = \mathbf{x}_0^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$$

Therefore

$$\begin{aligned} \text{Var}(\hat{\mu}_0 - \mu_0) &= \mathbf{x}_0^\top \text{Var}(\hat{\boldsymbol{\beta}}) \mathbf{x}_0 \\ &= \sigma^2 \mathbf{x}_0^\top (X^\top X)^{-1} \mathbf{x}_0 \end{aligned}$$

Result 2.3.4 (Confidence Interval for μ_0)

$$\hat{\mu}_0 \pm t_{n-p, 1-\alpha/2} s \sqrt{\mathbf{x}_0^\top (X^\top X)^{-1} \mathbf{x}_0}$$

Prediction Intervals

Suppose we observe a new predictor vector $\mathbf{x}_0$ and we want to predict a *future response*

$$Y_0 = \mathbf{x}_0^\top \boldsymbol{\beta} + \varepsilon_0$$

Unlike the mean response

$$\mu_0 := E[Y_0 | \mathbf{x}_0] = \mathbf{x}_0^\top \boldsymbol{\beta},$$

the future response Y_0 is itself random.

Definition 2.21 (Prediction Interval)

A $(1 - \alpha)$ prediction interval is a random interval

$$P(\mathbf{y}, \mathbf{x}_0) = [L(\mathbf{y}, \mathbf{x}_0), U(\mathbf{y}, \mathbf{x}_0)]$$

constructed so that

$$P\{Y_0 \in P(\mathbf{y}, \mathbf{x}_0)\} = 1 - \alpha$$

There are now two sources of uncertainty: uncertainty about $\boldsymbol{\beta}$ (estimation uncertainty) and ε_0 (irreducible future noise).

A future response at $\mathbf{x}_0$ is

$$Y_0 = \mathbf{x}_0^\top \boldsymbol{\beta} + \varepsilon_0, \quad \varepsilon_0 \perp\!\!\!\perp (\mathbf{y}, \hat{\boldsymbol{\beta}}), \quad \text{Var}(\varepsilon_0) = \sigma^2$$

Its prediction error is

$$Y_0 - \hat{\mu}_0 = \varepsilon_0 - \mathbf{x}_0^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$$

The two terms are independent, so

$$\begin{aligned} \text{Var}(Y_0 - \hat{\mu}_0) &= \text{Var}(\varepsilon_0) + \text{Var}\{\mathbf{x}_0^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\} \\ &= \sigma^2 (1 + \mathbf{x}_0^\top (X^\top X)^{-1} \mathbf{x}_0) \end{aligned}$$

Result 2.3.5 (Prediction Interval for Y_0)

$$\hat{\mu}_0 \pm t_{n-p, 1-\alpha/2} s \sqrt{1 + \mathbf{x}_0^\top (X^\top X)^{-1} \mathbf{x}_0}$$

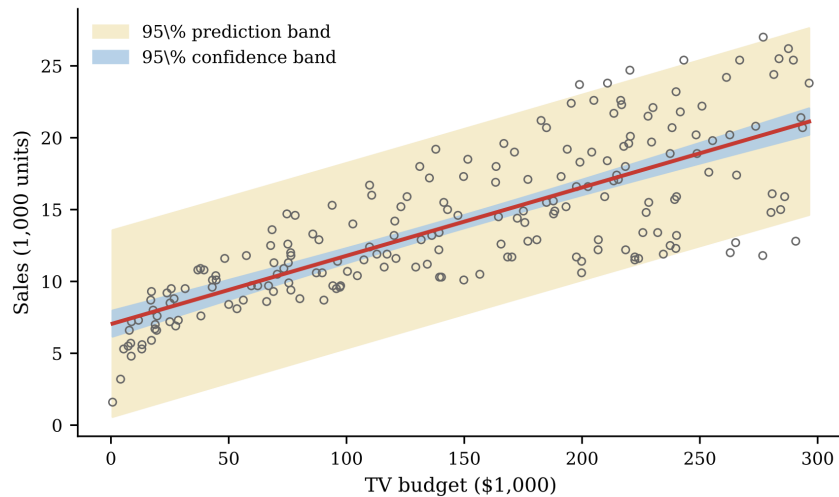
For the Advertising data (sales on TV and radio), at TV = \$100,000 and radio = \$20,000, both intervals are centred at $\hat{\mu}_0 = 11.256$, but

$$95\% \text{ CI for the mean response} = [10.985, 11.528],$$

whereas

$$95\% \text{ PI for a new response} = [7.930, 14.583]$$

The prediction interval is much wider because it includes both the uncertainty in estimating μ_0 (the $\mathbf{x}_0^\top (X^\top X)^{-1} \mathbf{x}_0$ term) and fresh outcome noise (the extra 1). The extra 1 is irreducible uncertainty about the future observation.



The confidence band is narrowest near the centre of the design, where $\mathbf{x}_0^\top (X^\top X)^{-1} \mathbf{x}_0$ is smallest. The prediction band is much wider because the irreducible σ^2 term dominates. Both formulas describe uncertainty under the fitted model; extrapolation far outside the observed predictor range is especially risky.

2.3.2 Testing a Single Coefficient (t-Test)

The Null Hypothesis for a Single Predictor

The t-Test

2.3.3 Testing Multiple Coefficients (F-Test)

The Global F-Test

The Partial F-Test

The Frisch-Waugh-Lovell Theorem

2.4 Extending the Linear Model

2.4.1 Qualitative Predictors

Dummy Variables and Baseline Coding

2.4.2 Interactions

Removing the Additivity Assumption

2.5 Regression Diagnostics

2.5.1 The Mean and Error Models

Diagnosing Non-linearity via Residual Plots

Correlated Errors

Heteroscedasticity and Weighted Least Squares

2.5.2 Unusual Observations

Outliers and Studentized Residuals

High-Leverage Points

2.5.3 Unstable Design and Non-Gaussian Errors

Collinearity and the Variance Inflation Factor

Checking Normality via Q-Q Plots

References